

AniruddhRaam Srinivasan

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[LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

Research Interests

Structural bioinformatics and material informatics, with an emphasis on Mixed-Solvent Molecular Dynamics (MSMD) and machine learning for cryptic pocket discovery. Decentralized multi-agent AI systems, edge-deployable computer vision, and secure, high-concurrency B2B software architecture.

Education

B.Tech, Artificial Intelligence & Data Science

Amrita Vishwa Vidyapeetham, Bengaluru, Karnataka, India

Aug 2024 – Present

CGPA: 8.44/10.00 | Prev. Sem. SGPA: 9.05/10.00

- **Core coursework:** Advanced Data Structures & Algorithm Analysis; Data Structures & Algorithms; Object Oriented Programming; Introduction to AI and Machine Learning; Big Data Analytics; Modeling, Simulation & Analysis; Software-Defined Communication Systems; Introduction to Computer Networks; Introduction to Robotics; Introduction to Material Informatics; Introduction to Biological Data; Computational Thinking; Elements of Computing; Introduction to Electrical and Electronics.

AISSCE (Grade 12)

Birla Open Minds International School

2024

76% (379/500)

AISSE (Grade 10)

Velammal Vidyalaya

2022

92% (459/500)

Publications

Peer-Reviewed Journal Articles

- Srinivasan, A., et al. "Generative Models for the Inverse Design and Band Gap Prediction of Semiconductor Materials." *Procedia Computer Science*, Elsevier, 2026.

Peer-Reviewed Conference Papers

- Srinivasan, A., Josheni, K. S., Vijay, S. "AI and robotics in education with focus in simulation." In *Information and Communication Systems*, Taylor & Francis, 2026.

Manuscripts Under Review / In Preparation

- Srinivasan, A., Madhulika, S., Venugopalan, M. "Edge-Enabled Multi-Agent Traffic Management System Using YOLO-Based Accident Detection and Adaptive Signal Control." 2026.

Honors, Awards & Competitive Achievements

- **University Award for High-Impact Digital Solutions, 2026:** Awarded by Amrita Vishwa Vidyapeetham for architecting the campus-wide project repository portal.
- **National Finalist (Top 6), Novo Nordisk Hackathon, 2025:** Selected as one of 6 teams for the national finals in a nationwide healthcare innovation hackathon, architecting an on-premise, secure-first API prototype using open source libraries and models.

Professional & Research Experience

Research Apprentice

IIIT Hyderabad

Jan 2026 – Apr 2026

- Spearheaded advanced bioinformatics research focused on the discovery of novel cryptic pockets for targeted cancer therapeutics.
- Executed and analyzed Mixed-Solvent Molecular Dynamics (MSMD) simulations utilizing GROMACS to identify and validate hidden binding sites in complex target proteins.

AI Engineer Intern

EdEasy (Hyderabad)

Dec 2025 – Jan 2026

- Designed complete end-to-end integration of AI endpoints (utilizing Gemini API) to process educational videos, providing automated summaries and content analysis.

- Implemented real-time dynamic question customization for students with AI-based response analysis to enhance personalized learning engagement.

Senior Executive

Nov 2025 – Jan 2026

Amrita Centre for Entrepreneurship (ACE)

- Coordinated large-scale event organization and managed comprehensive attendee coordination for entrepreneurship and networking events.

Research Projects

Generative Models for Semiconductor Inverse Design | *Python, XGBoost, GMM, NN*

Aug 2025 – Sep 2025

- Developed a hybrid generative framework using Gaussian Mixture Models (GMM) and Probability Mass Functions (PMF) to synthesize 5,000 physically consistent semiconductor materials.
- Engineered a machine learning pipeline utilizing an optimized XGBoost algorithm and Neural Networks for accurate band gap prediction, significantly accelerating materials discovery.

Agentic Cognitive Load Monitoring System | *Python, Audio Processing, AI Agents*

2026

- Engineered a real-time stress monitoring system for aviation pilots, utilizing AI agents to analyze critical voice acoustic components including jitter and shimmer.
- Designed automated cognitive load evaluation metrics to provide instantaneous feedback, enhancing situational awareness and operational safety in high-stakes environments.

Software, Systems & Engineering Projects

Amrita Project Portal | *React, Node.js, PostgreSQL, Docker*

Aug 2025 – Present

- Architected and deployed a full-stack portal with secure PostgreSQL in Docker, **supporting 2000+ students and 200+ faculty daily**. Developed under the collaborative guidance of Dr. Rajesh M and Dr. Sreevidya B.
- Engineered a project lifecycle module with secure JWT-based authentication, granular role-based authorization, and a Python/Flask API utilizing Regex for cumulative **SGPA and CGPA parsing, and student grading support from faculty**.

DBMS Architecture Analysis and Seminar: SAP ECC vs S4 HANA | *DBMS Coursework*

2026

- Researched and delivered a comprehensive seminar analyzing the architectural paradigm shift and query execution differences from SAP ECC to the S4 HANA database system, incorporating **real insights from company employees from Apple and IBM**.
- Evaluated and benchmarked the performance differentials between traditional row-based storage and modern columnar databases specifically for high-throughput analytics processing.

JavaFX Chess Engine Application | *Java, JavaFX, Stockfish API*

2026

- Developed a fully-featured, responsive desktop chess application integrating the Stockfish engine for scalable Human vs. AI gameplay across multiple configurable difficulty depths.
- Engineered rigorous move validation, check/checkmate detection algorithms, and dynamic UI thematic rendering utilizing robust object-oriented principles.

Hangman Jack (Nand2Tetris) | *Jack, VMEulator*

2026

- Engineered a terminal-based Hangman game in the Jack programming language, compiled and natively executed on the custom Nand2Tetris Hack VMEulator architecture.
- Implemented custom utility classes for memory-constrained I/O handling, dynamic heuristic scoring, and multi-player state management.

Linux Shell Replica | *C, Linux System Calls*

Dec 2025 – Feb 2026

- Developed a fully functional Linux shell replica in C, implementing native system calls for process management, execution, and I/O handling.

Level-5 IoT Air Quality Monitoring System | *ESP32, MQ135, PM2.5, JSON*

Apr 2026

- Designed a hardware-based air quality monitoring architecture with a structured JSON data pipeline streaming telemetry to the cloud.

Technical Skills

- **Programming Languages:** Python, Java, Go, C, JavaScript, SQL, Jack.
- **Machine Learning & AI:** LLMs (Ollama, Llama 3.1, Gemini), YOLOv11, ONNX, MLOps, Scikit-learn, OpenAI Whisper, Pandas, NumPy.
- **Systems, Web & Infrastructure:** React.js, Node.js, Flask, PostgreSQL, Docker, Tailwind CSS, Azure DevOps, Linux (Fedora, KDE Plasma).

- **Research Domains:** Bioinformatics (GROMACS, MSMD), Multi-Agent Systems, Edge Computer Vision, IoT Sensors, Voice Acoustics.

Professional & Transferable Skills

- **System Architecture & Strategy:** Proficient in orchestrating multi-year technical roadmaps, demonstrated by planning a 24-month B2B SaaS architecture for NBFC compliance orchestration.
- **Academic Collaboration:** Co-authored technical papers and developed enterprise-grade university platforms in tight collaboration with faculty and peers.
- **System Customization:** Advanced knowledge of high-performance Linux environments, dual-boot configurations, and system-wide bash scripting.

Certifications

- **Linux Administration**, Udemy (Apr 2026)
- **Generative AI in Action**, IBM (Feb 2026)
- **AI and ML with Python**, Next Chapter (Mar 2025)

Languages

English: Native | **Tamil:** Native | **Hindi:** Working Level Proficiency | **Swedish:** Beginner (Learning) | **German:** Beginner (Learning)

Interests

- 5+ Years of Learning Tabla. Active member of the university's Raaga Club, performing in multiple club events accompanying both Carnatic and Western music.
- High-performance computing Optimization and Linux desktop environments modifications and Linux Administration.
- Member of the college Athletics team and Professional Badminton player with 10+ years of playing.